

INTERNSHIP PROJECT REPORT

Sales Performance Analysis and Interactive Business Intelligence Dashboard

Data Analytics Internship

Submitted by
Manish Singh

Tools: Microsoft Excel | SQL | Python | NumPy | Power BI

Internship Submission Report

Internship Details

Applicant Name: Manish Singh

Application ID: KTS020260812721

Internship Domain: Data Analytics Internship

Duration: 30 Days

From: 20 August 2026

To: 19 September 2026

1. Project Overview

This internship project focuses on analyzing a real-world sales dataset and converting raw transactional data into meaningful business intelligence. The project follows an end-to-end analytics workflow covering data cleaning, database storage, statistical analysis, visualization, and business interpretation.

The cleaned data was prepared using Microsoft Excel and stored in a SQL database for structured and persistent data management. Python and NumPy were used for numerical and descriptive analysis, while Power BI was used to develop an interactive dashboard for monitoring sales, profit,

quantity, orders, customers, regional performance, product/category performance, and payment-method contribution.

2. Project Objectives

- Clean and preprocess the raw sales dataset.
- Store the cleaned dataset in a structured SQL database And Excel.
- Perform descriptive and numerical analysis using Python and NumPy.
- Identify yearly and monthly sales and profit trends.
- Evaluate category and product-level performance.
- Analyze regional sales and order performance.
- Understand sales-channel and customer purchasing behavior.
- Develop an interactive Power BI dashboard with relevant KPIs and visualizations.
- Generate actionable business insights and recommendations.

3. Dataset Description

Parameter	Value
Total Records	50,000
Total Sales	₹2,181,259,557
Total Quantity	149,676 units
Total Profit	₹194,163,285
Analysis Period	2023–2026
Regions	Central, East, North, South, West
Main Categories	Electronics, Furniture, Fashion, Home Appliances, Office Supplies
Product Types	15

The dataset contains transactional sales information with measures and dimensions such as Order_ID, date, region, category, product, quantity, discount, unit price, sales, profit, profit margin, payment method, sales channel, and customer-related information.

4. Tools & Technologies

Tool / Technology	Purpose
Microsoft Excel	Data cleaning, preprocessing, Pivot Tables and initial exploratory analysis
SQL	Structured storage and retrieval of cleaned data
Python	Data analysis and numerical processing

NumPy	Statistical and numerical analysis
Power BI	Interactive dashboard, KPI analysis and visualization
Jupyter Notebook	Python analysis and documentation
GitHub	Source code, notebook and project documentation

5. Data Cleaning & Preprocessing

The raw sales dataset was first inspected for data quality and consistency. Microsoft Excel was used to prepare the data for analysis. The cleaning stage focused on validating the dataset structure, checking duplicates and missing values, standardizing formats, and ensuring that numerical and categorical fields were appropriate for analysis.

5.1 Cleaning Workflow

- Inspect the raw dataset and identify data-quality issues.
- Check duplicate and missing records.
- Validate date, numerical and categorical fields.
- Standardize inconsistent values and formats where required.
- Remove or correct invalid records where applicable.
- Validate the cleaned dataset before analytical processing.

6. SQL Database Integration

After cleaning, the prepared dataset was stored in a SQL database. This provides a structured and persistent storage layer and makes the cleaned data available for future querying, analysis, reporting, and integration with analytics tools.

Workflow: Raw Data → Excel Cleaning → Clean Dataset → SQL Database → Python / Power BI → Insights

SQL was used as the database layer for maintaining the cleaned dataset. The database can also support future extensions such as automated data pipelines, scheduled refreshes, and SQL-based analytical queries.

7. Exploratory Data Analysis

Pivot-based analysis was performed to understand sales, profit, orders, categories, products, regions, and channels before developing the final dashboard.

7.1 Year-wise Sales Analysis

Year	Sales
2023	₹594.73M
2024	₹607.94M
2025	₹608.13M
2026	₹370.46M

Sales increased from 2023 to 2025, with 2025 recording approximately ₹608.13M. The 2026 value should be interpreted according to the available reporting period; if it represents a partial year, it should not be compared directly with complete years.

7.2 Year-wise Profit Analysis

Year	Profit
2023	₹53.10M
2024	₹53.98M
2025	₹54.00M
2026	₹33.09M

7.3 Monthly Sales Analysis

Mon th	Sales
Jan	₹203.91M
Feb	₹190.11M
Mar	₹196.49M
Apr	₹210.30M
May	₹198.79M
Jun	₹193.09M
Jul	₹194.19M
Aug	₹185.30M
Sep	₹141.69M
Oct	₹152.82M
Nov	₹160.81M
Dec	₹153.76M

April recorded the highest monthly sales at approximately ₹210.30M, while September recorded the lowest at approximately ₹141.69M. The pattern indicates stronger sales in the first and middle portions of the year and a noticeable decline from September onward.

7.4 Monthly Profit Analysis

Month	Profit
Jan	₹18.35M
Feb	₹17.12M
Mar	₹17.53M
Apr	₹18.59M
May	₹17.73M
Jun	₹17.52M
Jul	₹17.39M
Aug	₹16.20M
Sep	₹12.45M
Oct	₹13.56M
Nov	₹13.87M
Dec	₹13.85M

April generated the highest monthly profit at approximately ₹18.59M, while September recorded the lowest at approximately ₹12.45M. Profit broadly follows the sales pattern.

8. Category Performance Analysis

Category	Sales	Profit	Orders
Electronics	₹1,659.46M	₹128.33M	17,992
Furniture	₹212.08M	₹26.28M	9,976
Home Appliances	₹175.98M	₹18.42M	8,052
Office Supplies	₹101.72M	₹16.54M	7,866
Fashion	₹32.02M	₹4.59M	6,114

Electronics is the dominant category, contributing approximately **₹1.66B** in sales and **₹128.33M** in profit. It also records the highest order volume with **17,992** orders. This makes Electronics a critical category for inventory, pricing, promotion, and profitability decisions.

9. Product Performance Analysis

Product	Sales
Laptops	₹962.07M
Mobiles	₹657.26M
Tables	₹98.48M
Printing	₹96.15M
Storage	₹67.57M

Laptops are the highest-selling product at approximately **₹962.07M**, followed by Mobiles at approximately **₹657.26M**. These products represent major revenue contributors and should receive appropriate attention in inventory and sales planning.

10. Regional Sales Analysis

Region	Sales	Orders
Central	₹275.26M	6,356
East	₹367.18M	8,440
North	₹632.19M	14,399
South	₹459.68M	10,532
West	₹446.96M	10,273

North is the strongest region, generating approximately **₹632.19M** in sales and **14,399** orders. South and West also show substantial contribution, while Central has the lowest sales and order volume among the five regions.

11. Sales Channel & Customer Purchasing Analysis

The dataset includes Marketplace, Online, and Retail Store sales channels. Channel-level order analysis was used to understand purchasing activity and identify differences in customer behavior across sales channels.

The Power BI dashboard further supports analysis through interactive filtering and payment method visualization, allowing business users to explore how customers transact and how payment methods contribute to profit.

12. Python & NumPy Analysis

Python was used to perform numerical analysis and validate key business metrics. NumPy-based analysis and descriptive statistics provided a quantitative view of the dataset.

Metric	Result
Total Sales	₹2,181,259,557.65
Total Profit	₹191,549,820.35
Total Quantity	149,676
Average Quantity	2.99
Average Discount	0.102
Average Unit Price	₹16,139.24
Average Sales per Record	₹43,625.19
Average Profit per Record	₹3,830.996
Average Profit Margin	0.10877

12.1 Descriptive Statistics

Variable	Min	Median	Max
Quantity	1	2	10
Discount	0	0.10	0.30
Unit Price	₹162.04	₹6,955.83	₹71,497.92
Sales	₹120.71	₹13,728.88	₹711,167.90
Profit	-₹17,705.84	₹1,265.34	₹104,456.85
Profit Margin	-0.0419	0.1087	0.2638

The minimum profit is negative, indicating that the dataset contains loss-making transactions. The maximum profit is approximately ₹104,456.85, demonstrating substantial variation in transaction-level profitability.

13. Power BI Dashboard Development

An interactive Power BI dashboard was developed to convert the analytical results into a business-friendly visual reporting interface. The dashboard combines KPI cards, trend charts, regional analysis, category analysis, and payment-method analysis.

13.1 Dashboard KPIs

KPI	Value
Total Sales	₹2,181.26M
Total Profit	₹194.55M
Total Quantity	149.68K
Profit Margin	10.88%
Total Orders	Verified dashboard value
Total Customers	Verified dashboard value

The KPI values should reflect the final corrected Power BI model after the reconciliation of aggregation logic and data refresh. Total profit and order count were verified/fixed before final submission.

13.2 Dashboard Visualizations

- Monthly Sales – identifies monthly revenue trends.
- Monthly Profit – tracks monthly profitability.
- Sales Region Wise – compares sales-channel performance across regions.
- Profit Region Wise – compares profitability across regions.
- Category Wise Sales – identifies major revenue-generating categories.
- Profit by Payment Method – compares profit contribution across payment methods.
- KPI Cards – provide a quick summary of core business metrics.

14. Key Business Insights

- The dataset contains 50,000 transactional records and generates approximately ₹2.18B in total sales.
- Electronics is the strongest category, contributing approximately ₹1.66B in sales and ₹128.33M in profit.
- Laptops are the highest-selling product at approximately ₹962.07M, followed by Mobiles at approximately ₹657.26M.
- North is the highest-performing region with approximately ₹632.19M in sales and 14,399 orders.

- April recorded the highest monthly sales at approximately ₹210.30M and the highest monthly profit at approximately ₹18.59M.
- September recorded the lowest monthly sales at approximately ₹141.69M and the lowest monthly profit at approximately ₹12.45M.
- The presence of negative-profit transactions indicates that some sales are loss-making and may require further investigation.
- Customer purchasing activity can be evaluated through sales channels and payment methods available in the dashboard.

15. Business Recommendations

- Prioritize inventory availability and sales planning for Electronics, Laptops, and Mobiles because of their strong revenue contribution.
- Investigate loss-making transactions to identify whether discounts, pricing, procurement cost, or other factors are reducing profitability.
- Analyze the September sales decline and consider targeted promotions or demand-generation strategies during weaker months.
- Use the North region as a benchmark for identifying practices that could be applied to lowerperforming regions.
- Develop category-specific marketing strategies for lower-performing categories such as Fashion.
- Use sales-channel and payment-method insights to optimize channel-specific customer engagement and transaction strategies.
- Use monthly sales and profit trends to improve inventory planning, staffing, and promotional scheduling.

16. Challenges & Solutions

Challenge	Approach / Solution
Large transactional dataset	Used Pivot Tables and programmatic analysis for aggregation.
Data quality and consistency	Performed cleaning and validation before analysis.
Multiple business dimensions	Analyzed time, category, product, region and channel dimensions separately.
Need for structured storage	Stored the cleaned dataset in SQL.
Complex business information	Converted analytical results into interactive Power BI visuals.
Finding actionable patterns	Compared sales, profit, orders and product/region performance.

17. End-to-End Project Workflow

Raw Sales Data → Excel Data Cleaning → Cleaned Dataset → SQL Database → Python/NumPy Analysis → Power BI Data Model & Dashboard → Business Insights → Recommendations

18. Learning Outcomes

- Practical experience in data cleaning and preprocessing.
- Hands-on use of Excel Pivot Tables for business analysis.
- Understanding of SQL-based data storage and retrieval.
- Practical Python and NumPy experience for numerical analysis.
- Experience in developing KPI-driven Power BI dashboards.
- Improved understanding of data visualization and business intelligence.
- Ability to convert analytical results into business recommendations.

19. Conclusion

The Sales Performance Analysis project demonstrates an end-to-end data analytics workflow from raw data preparation to business intelligence reporting. The dataset was cleaned using Excel, stored in SQL, analyzed using Python and NumPy, and transformed into an interactive Power BI dashboard. The analysis identified key sales and profit trends, leading product and category contributors, regional performance differences, and customer purchasing dimensions. The resulting insights can support inventory planning, product prioritization, regional strategy, promotional planning, and data-driven decision making.

20. Future Scope

- Integrate the SQL database directly with Power BI for a more automated reporting pipeline.
- Implement scheduled data refresh and automated dashboard updates.
- Develop sales and profit forecasting models.
- Perform customer segmentation and customer lifetime value analysis.
- Build predictive models for sales and profitability.
- Introduce automated business alerts for unusual sales or profit movements.

21. GitHub Repository & Deliverables

The final project package can include the public GitHub repository, Jupyter Notebook/source code, Power BI dashboard file, README documentation, and this Business Insights Report. The repository URL should be inserted below before submission:

GitHub Repository: [github Link](#)

Dashboard File: SALES_DATA_POWER_BI

Numpy Notebook(File Name): CDA

22. Appendix

Appendix A — Excel Data Cleaning

Order ID	Order Date	Order Time	Customer ID	Customer Name	Category	Sub-Category	Product Name	Region	State	City	Quantity	Discount	Unit Price	Sales	Profit	Payment Method	Customer Segment	Sales Channel	Tran	Quart	Month
ORD1058101-01-2023	00:16:58	CUST11884	Arjun Bansal	Electronics	Accessories	Power Bank 20000mAh	East	Bihar	Patna	1	0.1	1680.68	1512.61	109.21	Credit Card	Consumer	Online	2023 Q1	Jan	2	
ORD1304241-01-01-2023	00:23:28	CUST17134	Sneha Agarwal	Electronics	Accessories	Power Bank 20000mAh	South	Telangana	Hyderabad	4	0.2	1669.86	5343.55	109.54	UPI	Corporate	Retail Store	2023 Q1	Jan	2	
ORD1444801-01-01-2023	00:46:50	CUST13701	Nandini Verma	Furniture	Storage	Bookshelf	North	Uttarakhand	Dehradun	8	0.05	5458.54	41561.66	4929.42	Net Banking	Small Business	Online	2023 Q1	Jan	2	
ORD142941-01-01-2023	00:52:17	CUST15691	Manish Pandey	Furniture	Storage	Filing Cabinet	East	Odisha	Bhubaneswar	2	0	6929.25	13858.5	2138.52	Debit Card	Corporate	Online	2023 Q1	Jan	2	
ORD1268401-01-01-2023	01:15:51	CUST14335	Neha Malhotra	Electronics	Mobiles	iPhone 15	South	Karnataka	Bengaluru	1	0.05	69925.84	66429.55	7446.75	UPI	Consumer	Retail Store	2023 Q1	Jan	2	
ORD1223591-01-01-2023	01:32:11	CUST15049	Sneha Das	Electronics	Laptops	HP Pavilion 15	West	Goa	Panaji	2	0	62740.95	125481.9	15095.47	Net Banking	Corporate	Online	2023 Q1	Jan	2	
ORD1300951-01-01-2023	03:00:14	CUST16003	Raj Verma	Electronics	Mobiles	OnePlus Nord CE	West	Goa	Panaji	4	0.15	29902.69	101669.2	4026.10	Debit Card	Consumer	Online	2023 Q1	Jan	2	
ORD1490041-01-01-2023	03:52:37	CUST17928	Rahul Kumar	Fashion	Footwear	Running Shoes	North	Punjab	Ludhiana	3	0	3458.72	10376.16	1656.01	Net Banking	Corporate	Retail Store	2023 Q1	Jan	2	
ORD11619101-01-01-2023	03:54:18	CUST10899	Rahul Mehta	Furniture	Tables	Study Table	East	Jharkhand	Ranchi	1	0.15	4397.23	5437.65	400.93	EMI	Consumer	Marketplace	2023 Q1	Jan	2	
ORD126491-01-01-2023	04:49:32	CUST16374	Simran Agarwal	Furniture	Chairs	Ergonomic Office Chair	West	Gujarat	Ahmedabad	2	0.15	8897.18	15125.21	1427.76	Credit Card	Consumer	Online	2023 Q1	Jan	2	
ORD134591-01-01-2023	05:20:07	CUST13140	Vivaan Bansal	Office Supplies	Stationery	Premium Notebook	East	Bihar	Patna	1	0.15	484.83	412.11	60.53	Cash	Consumer	Online	2023 Q1	Jan	2	
ORD1393171-01-01-2023	05:40:27	CUST16915	Raj Choudhary	Home Appliances	Kitchen	Air Fryer	West	Goa	Panaji	4	0.2	6762.55	21640.16	951.99	Debit Card	Consumer	Retail Store	2023 Q1	Jan	2	
ORD1145461-01-01-2023	06:50:30	CUST17226	Simran Sharma	Fashion	Footwear	Running Shoes	East	West Bengal	Kolkata	3	0	3300.5	9901.5	2354.88	Debit Card	Corporate	Online	2023 Q1	Jan	2	
ORD14445101-01-01-2023	07:20:52	CUST12844	Neha Mishra	Home Appliances	Kitchen	Air Fryer	West	Maharashtra	Mumbai	10	0.05	7817.35	74264.82	7816.13	Debit Card	Small Business	Online	2023 Q1	Jan	2	
ORD1434041-01-01-2023	08:58:37	CUST15409	Ananya Joshi	Furniture	Tables	Office Work Desk	South	Telangana	Hyderabad	1	0	12395.31	12395.31	2513.53	Credit Card	Consumer	Marketplace	2023 Q1	Jan	2	
ORD124231-01-01-2023	09:05:14	CUST15975	Pooja Bansal	Electronics	Laptops	HP Pavilion 15	East	Bihar	Patna	1	0.05	56756.51	53918.68	6491.81	Debit Card	Consumer	Online	2023 Q1	Jan	2	
ORD105291-01-01-2023	09:54:13	CUST11506	Rohan Joshi	Fashion	Women	Women's Kurti	West	Maharashtra	Mumbai	2	0	1242.97	2485.94	450.48	Cash	Consumer	Marketplace	2023 Q1	Jan	2	
ORD1431481-01-01-2023	10:00:14	CUST14352	Priya Mehta	Home Appliances	Kitchen	Air Fryer	South	Andhra Pradesh	Vijayawada	1	0.15	7031.7	5976.94	373.65	Debit Card	Consumer	Online	2023 Q1	Jan	2	
ORD1381031-01-01-2023	10:22:18	CUST15812	Riya Sharma	Electronics	Accessories	Bluetooth Speaker	South	Kerala	Kochi	1	0.2	3741.49	2993.19	50.58	UPI	Corporate	Online	2023 Q1	Jan	2	
ORD1358621-01-01-2023	10:27:12	CUST14895	Vivaan Sharma	Office Supplies	Stationery	Ball Pen Pack	West	Gujarat	Surat	2	0.1	189.86	341.75	60.83	UPI	Consumer	Retail Store	2023 Q1	Jan	2	
ORD1370481-01-01-2023	10:45:29	CUST14483	Manish Malhotra	Electronics	Accessories	Wireless Earbuds	North	Uttar Pradesh	Kanpur	4	0.3	2188.6	6128.08	207.74	Credit Card	Consumer	Online	2023 Q1	Jan	2	
ORD130461-01-01-2023	13:33:14	CUST14149	Priya Malhotra	Electronics	Laptops	Lenovo IdeaPad Slim	West	Gujarat	Surat	3	0.05	56127.42	159963.2	13484.89	Credit Card	Consumer	Retail Store	2023 Q1	Jan	2	
ORD103041-01-01-2023	13:33:34	CUST17122	Neha Kumar	Office Supplies	Paper	A4 Paper Ream	North	Uttar Pradesh	Kanpur	2	0.05	378.71	719.55	150.09	UPI	Consumer	Online	2023 Q1	Jan	2	
ORD139501-01-01-2023	15:34:54	CUST13836	Arjun Choudhary	Electronics	Accessories	Bluetooth Speaker	South	Tamil Nadu	Chennai	5	0	3540.61	17703.05	1975.66	Credit Card	Consumer	Retail Store	2023 Q1	Jan	2	
ORD104591-01-01-2023	15:46:23	CUST15644	Priya Joshi	Home Appliances	Kitchen	Air Fryer	South	Andhra Pradesh	Vijayawada	4	0.1	4765.2	24364.72	2534.15	Net Banking	Consumer	Marketplace	2023 Q1	Jan	2	
ORD1078401-01-01-2023	17:06:42	CUST13207	Asha Jain	Office Supplies	Printing	Laser Printer	North	Uttarakhand	Dehradun	4	0.15	1983.95	67938.63	10866.94	Debit Card	Small Business	Marketplace	2023 Q1	Jan	2	
ORD1011941-01-01-2023	17:07:42	CUST11188	Pooja Bansal	Furniture	Chairs	Ergonomic Office Chair	South	Tamil Nadu	Chennai	6	0.25	7840.44	35281.89	1349.09	Cash	Consumer	Online	2023 Q1	Jan	2	
ORD101231-01-01-2023	18:30:56	CUST16111	Ankit Kumar	Electronics	Laptops	Dell Inspiron 15	South	Telangana	Hyderabad	1	0.15	61145.54	51973.71	2395.99	Debit Card	Consumer	Retail Store	2023 Q1	Jan	2	
ORD137751-01-01-2023	20:05:43	CUST16767	Pooja Srivastava	Fashion	Men	Men's Casual Shirt	South	Kerala	Kochi	4	0.25	1353.6	4060.8	241.21	EMI	Corporate	Online	2023 Q1	Jan	2	
ORD1270401-01-01-2023	21:09:39	CUST12432	Manish Mishra	Office Supplies	Paper	A4 Paper Ream	East	Odisha	Bhubaneswar	5	0.1	410.64	1847.88	283.38	Credit Card	Corporate	Online	2023 Q1	Jan	2	
ORD1311461-01-01-2023	21:38:25	CUST11003	Nandini Yadav	Furniture	Tables	Study Table	North	Punjab	Ludhiana	2	0.15	6655.64	11514.59	919.09	Debit Card	Small Business	Marketplace	2023 Q1	Jan	2	
ORD1300301-01-01-2023	22:16:46	CUST10841	Ayush Kumar	Fashion	Women	Women's Kurti	South	Telangana	Hyderabad	6	0.1	1276.22	6891.59	1153.29	Net Banking	Consumer	Online	2023 Q1	Jan	2	
ORD100141-01-01-2023	22:35:37	CUST15331	Meera Verma	Electronics	Accessories	Bluetooth Speaker	West	Goa	Panaji	3	0.2	3666.33	8799.19	147.83	Cash	Small Business	Marketplace	2023 Q1	Jan	2	
ORD147231-01-01-2023	22:49:20	CUST15331	Aditi Srivastava	Electronics	Laptops	Lenovo IdeaPad Slim	South	Tamil Nadu	Chennai	3	0.1	48160.12	130032.3	8751.18	Credit Card	Consumer	Retail Store	2023 Q1	Jan	2	
ORD130731-01-01-2023	23:35:31	CUST11677	Nandini Shah	Furniture	Chairs	Ergonomic Office Chair	Central	Madhya Pradesh	Bhopal	10	0.25	8282.93	62121.98	2357.08	Debit Card	Consumer	Retail Store	2023 Q1	Jan	2	
ORD138061-01-01-2023	23:40:51	CUST14106	Ishita Srivastava	Fashion	Footwear	Running Shoes	Central	Chhattisgarh	Raipur	4	0.15	3027.76	10294.38	1313.27	UPI	Consumer	Online	2023 Q1	Jan	2	
ORD11972102-01-01-2023	00:06:34	CUST14077	Arjun Das	Home Appliances	Kitchen	Mixer Grinder	North	Uttar Pradesh	Kanpur	7	0	4139.24	28974.68	4048.07	UPI	Consumer	Retail Store	2023 Q1	Jan	2	
ORD135641-02-01-2023	00:26:28	CUST16984	Kavya Singh	Electronics	Laptops	HP Pavilion 15	North	Uttar Pradesh	Kanpur	4	0.1	65542.57	235953.3	15030.22	Credit Card	Consumer	Retail Store	2023 Q1	Jan	2	
ORD136641-02-01-2023	00:33:34	CUST12384	Nandini Malhotra	Fashion	Footwear	Running Shoes	South	Karnataka	Bengaluru	3	0.1	3444.93	9301.31	1054.48	Net Banking	Consumer	Online	2023 Q1	Jan	2	
ORD10997102-01-01-2023	01:09:01	CUST12849	Simran Verma	Electronics	Laptops	Lenovo IdeaPad Slim	North	Haryana	Gurgaon	2	0	53521.63	107043.3	16441.84	Debit Card	Consumer	Retail Store	2023 Q1	Jan	2	
ORD113869102-01-01-2023	01:11:11	CUST11263	Vivaan Bansal	Electronics	Laptops	Lenovo IdeaPad Slim	North	Haryana	Gurgaon	3	0.1	56580.32	152766.9	12710.20	Cash	Corporate	Online	2023 Q1	Jan	2	

Excel Clean Data

Appendix B — SQL Database

pgAdmin 4

Sales_Data/postgres@PostgreSQL 18*

Sales_Data/postgres@PostgreSQL 18

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 50

Order_ID	Order_Date	Customer_ID	Customer_Name	Category	Sub_Category	Product_Name	Region	State	City	Quantity	Discount	Unit_Pri
1	ORD1058...	2023-01-01	CUST11884	Arjun Bansal	Electronics	Accessories	Power Bank 20000m...	East	Bihar	Patna	1	0.1
2	ORD1304...	2023-01-01	CUST117134	Sneha Agarwal	Electronics	Accessories	Power Bank 20000m...	South	Telangana	Hyderabad	4	0.2
3	ORD1444...	2023-01-01	CUST113701	Nandini Verma	Furniture	Storage	Bookshelf	North	Uttarakhand	Dehradun	8	0.05
4	ORD1429...	2023-01-01	CUST115691	Manish Pandey	Furniture	Storage	Filing Cabinet	East	Odisha	Bhubanesw...	2	0
5	ORD1268...	2023-01-01	CUST14335	Neha Malhotra	Electronics	Mobiles	iPhone 15	South	Karnataka	Bengaluru	1	0.05
6	ORD1223...	2023-01-01	CUST15049	Sneha Das	Electronics	Laptops	HP Pavilion 15	West	Goa	Panaji	2	0
7	ORD1300...	2023-01-01	CUST16003	Raj Verma	Electronics	Mobiles	OnePlus Nord CE	West	Goa	Panaji	4	0.15
8	ORD1490...	2023-01-01	CUST17928	Rahul Kumar	Fashion	Footwear	Running Shoes	North	Punjab	Ludhiana	3	0
9	ORD1161...	2023-01-01	CUST10899	Rahul Mehta	Furniture	Tables	Study Table	East	Jharkhand	Ranchi	1	0.15
10	ORD1264...	2023-01-01	CUST16374	Simran Agarwal	Furniture	Chairs	Ergonomic Office Ch...	West	Gujarat	Ahmedabad	2	0.15
11	ORD1345...	2023-01-01	CUST113140	Vivaan Bansal	Office Suppli...	Stationery	Premium Notebook	East	Bihar	Patna	1	0.15
12	ORD1393...	2023-01-01	CUST16915	Raj Choudhary	Home Applia...	Kitchen	Air Fryer	West	Goa	Panaji	4	0.2
13	ORD1145...	2023-01-01	CUST17226	Simran Sharma	Fashion	Footwear	Running Shoes	East	West Bengal	Kolkata	3	0
14	ORD1444...	2023-01-01	CUST12944	Neha Mishra	Home Applia...	Kitchen	Air Fryer	West	Maharashtra	Mumbai	10	0.05
15	ORD1434...	2023-01-01	CUST15409	Ananya Joshi	Furniture	Tables	Office Work Desk	South	Telangana	Hyderabad	1	0
16	ORD1242...	2023-01-01	CUST15975	Pooja Bansal	Electronics	Laptops	HP Pavilion 15	East	Bihar	Patna	1	0.05
17	ORD1052...	2023-01-01	CUST11506	Rohan Joshi	Fashion	Women's	Women's Kurli	West	Maharashtra	Mumbai	2	0
18	ORD1431...	2023-01-01	CUST14352	Priya Mehta	Home Applia...	Kitchen	Air Fryer	South	Andhra Prade...	Vijayawada	1	0.15
19	ORD1381...	2023-01-01	CUST15812	Riya Sharma	Electronics	Accessories	Bluetooth Speaker	South	Kerala	Kochi	1	0.2
20	ORD1358...	2023-01-01	CUST14895	Vivaan Sharma	Office Suppli...	Stationery	Ball Pen Pack	West	Gujarat	Surat	2	0.1
21	ORD1370...	2023-01-01	CUST14483	Manish Malhotra	Electronics	Accessories	Wireless Earbuds	North	Uttar Pradesh	Kanpur	4	0.3
22	ORD1336...	2023-01-01	CUST14149	Priya Malhotra	Electronics	Laptops	Lenovo IdeaPad Slim	West	Gujarat	Surat	3	0.05
23	ORD1030...	2023-01-01	CUST17122	Neha Kumar	Office Suppli...	Paper	A4 Paper Ream	North	Uttar Pradesh	Kanpur	2	0.05

Total rows: 50000 Query complete 00:00:00.477

Cleaned Sales Dataset Stored in SQL Database

Appendix C — Python & NumPy Analysis



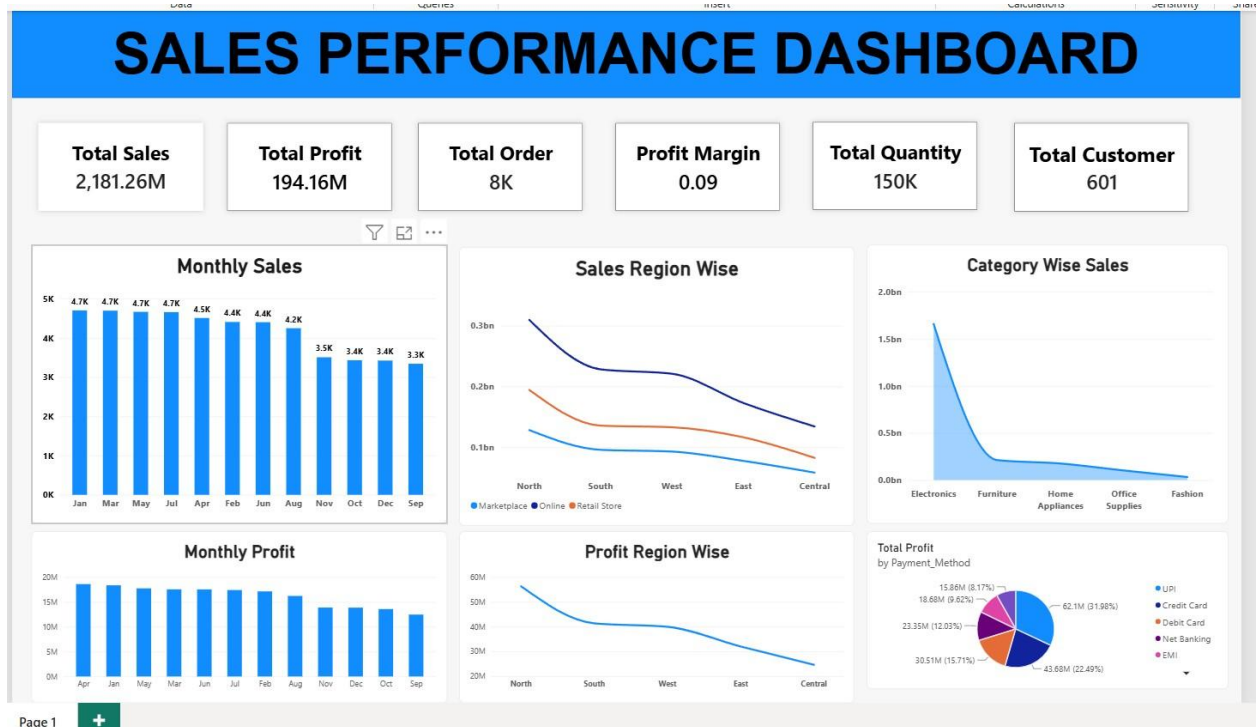
Python-Based Sales Analysis

<pre>data.describe(include="object")</pre>															
	Order_ID	Customer_ID	Customer_Name	Category	Sub_Category	Product_Name	Region	State	City	Payment_Method	Customer_Segment	Sales_Channel	Quarter	Month	Year
count	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000
unique	50000	7985	600	5	15	25	5	20	24	6	3	3	4	12	
top	ORD116464	CUST16756	Shreya Singh	Electronics	Laptops	HP Pavilion 15	North	Madhya Pradesh	Raipur	UPI	Consumer	Online	Q1	Jan	2
freq	1	18	124	17992	6079	2074	14399	4177	2179	16044	27657	24955	13807	4703	

<pre>data.describe(include="all")</pre>															
	Order_ID	Order_Date	Customer_ID	Customer_Name	Category	Sub_Category	Product_Name	Region	State	City	Quantity	Discount	Unit_Price	Sales	
count	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000.000000	50000.000000	50000.000000	50000.000000	
unique	50000	NaN	7985	600	5	15	25	5	20	24	NaN	NaN	NaN	NaN	
top	ORD116464	NaN	CUST16756	Shreya Singh	Electronics	Laptops	HP Pavilion 15	North	Madhya Pradesh	Raipur	NaN	NaN	NaN	NaN	
freq	1	NaN	18	124	17992	6079	2074	14399	4177	2179	NaN	NaN	NaN	NaN	
mean	NaN	2024-10-25 18:24:48.073440256	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2.993520	0.101938	16139.241250	43625.191153	
min	NaN	2023-01-01 00:16:58	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1.000000	0.000000	162.040000	120.710000	
25%	NaN	2023-11-26 11:33:51.500000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1.000000	0.050000	2102.390000	4124.057500	
50%	NaN	2024-10-28 21:40:10	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2.000000	0.100000	6955.830000	13728.880000	

Python-Based Sales Analysis

Appendix D — Power BI Dashboard



Sales Performance Power BI Dashboard